Assignment 4: Project 3 Presentation

Prediction Model Findings + Programming Concepts

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Insights of House Price Driving Factors

Project Overview

- **Dataset:** The Ames Housing.
- **Prediction goal:** Find the influential factors that drives sales price.
- **Audience:** Important for property investors and owners.

Problem Statement

What problem am I solving?

- The problem is to predict sales prices by analyzing influential factors such as overall quality, neighborhood, lot area, Gross Living Area, garage car.
- These models help having a accurate price prediction which helps buys, sellers, and investment agents to make informed financial decision by not overpaying and underpaying anyone.
- These models uses the provided historical data to make accurate prediction. By comparing different models, we can determine which variables influences the house price more.

Audience

Who is this presentation for?

- This presentation is for house buyers and sellers, and property investors.
- They can use the insights to make decision on house prices, evaluating house prices, or identifying the influential factors.
- The audience doesn’t need any deep technical knowledge. They need basic understanding of graphs and regression line.

Setup

```
# Load Libraries
library(ggplot2) # This library is useful for data visualization
library(scales) #Formats output data
library(janitor) #Formats output data
library(tidyverse)
library(forcats)
library(rlang)
library(patchwork)
library(gt)
library(glmnet)
library(tidymodels)  # modern unified framework (includes rsample, parsnip, ya
library(broom)       # tidy model summaries
library(performance) # model diagnostics
library(see)         # visualization for performance package
library(vip)         # variable importance plots
library(MASS)
library(car)        # for VIF
library(caret)      # for preprocessing and modeling
library(dplyr) # This library is useful for data manipulation
library(conflicted)

library(tidyverse)

library(AmesHousing)
bush_data <- make_ames()
```

Data Overview

What data am I using?

```
# View the structure of the dataset
glimpse(bush_data)
```

```
Rows: 2,930
Columns: 81
$ MS_SubClass        <fct> One_Story_1946_and_Newer_All_Styles, One_Story_1946…
$ MS_Zoning          <fct> Residential_Low_Density, Residential_High_Density, …
$ Lot_Frontage       <dbl> 141, 80, 81, 93, 74, 78, 41, 43, 39, 60, 75, 0, 63,…
$ Lot_Area           <int> 31770, 11622, 14267, 11160, 13830, 9978, 4920, 5005…
$ Street             <fct> Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pav…
$ Alley              <fct> No_Alley_Access, No_Alley_Access, No_Alley_Access, …
$ Lot_Shape          <fct> Slightly_Irregular, Regular, Slightly_Irregular, Re…
$ Land_Contour       <fct> Lvl, Lvl, Lvl, Lvl, Lvl, Lvl, Lvl, HLS, Lvl, Lvl, L…
$ Utilities          <fct> AllPub, AllPub, AllPub, AllPub, AllPub, AllPub, All…
$ Lot_Config         <fct> Corner, Inside, Corner, Corner, Inside, Inside, Ins…
$ Land_Slope         <fct> Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, G…
$ Neighborhood       <fct> North_Ames, North_Ames, North_Ames, North_Ames, Gil…
$ Condition_1        <fct> Norm, Feedr, Norm, Norm, Norm, Norm, Norm, Norm, No…
$ Condition_2        <fct> Norm, Norm, Norm, Norm, Norm, Norm, Norm, Norm, Nor…
$ Bldg_Type          <fct> OneFam, OneFam, OneFam, OneFam, OneFam, OneFam, Twn…
$ House_Style        <fct> One_Story, One_Story, One_Story, One_Story, Two_Sto…
$ Overall_Qual       <fct> Above_Average, Average, Above_Average, Good, Averag…
$ Overall_Cond       <fct> Average, Above_Average, Above_Average, Average, Ave…
$ Year_Built         <int> 1960, 1961, 1958, 1968, 1997, 1998, 2001, 1992, 199…
$ Year_Remod_Add     <int> 1960, 1961, 1958, 1968, 1998, 1998, 2001, 1992, 199…
```

```
# Show a few rows
head(bush_data)
```

```
# A tibble: 6 × 81
  MS_SubClass             MS_Zoning Lot_Frontage Lot_Area Street Alley Lot_Shape
  <fct>                   <fct>            <dbl>    <int> <fct>  <fct> <fct>
1 One_Story_1946_and_New… Resident…          141    31770 Pave   No_A… Slightly…
2 One_Story_1946_and_New… Resident…           80    11622 Pave   No_A… Regular
3 One_Story_1946_and_New… Resident…           81    14267 Pave   No_A… Slightly…
4 One_Story_1946_and_New… Resident…           93    11160 Pave   No_A… Regular
5 Two_Story_1946_and_New… Resident…           74    13830 Pave   No_A… Slightly…
6 Two_Story_1946_and_New… Resident…           78     9978 Pave   No_A… Slightly…
```

```
# ℹ 74 more variables: Land_Contour <fct>, Utilities <fct>, Lot_Config <fct>,
#   Land_Slope <fct>, Neighborhood <fct>, Condition_1 <fct>, Condition_2 <fct>,
#   Bldg_Type <fct>, House_Style <fct>, Overall_Qual <fct>, Overall_Cond <fct>,
#   Year_Built <int>, Year_Remod_Add <int>, Roof_Style <fct>, Roof_Matl <fct>,
#   Exterior_1st <fct>, Exterior_2nd <fct>, Mas_Vnr_Type <fct>,
#   Mas_Vnr_Area <dbl>, Exter_Qual <fct>, Exter_Cond <fct>, Foundation <fct>,
#   Bsmt_Qual <fct>, Bsmt_Cond <fct>, Bsmt_Exposure <fct>, …
```

Explanation:

- There are 2,930 rows and 81 columns in your dataset.
- The dataset represents the variables of houses in Ames in Iowa.
- Important data limitations can be not having all the factors such as current interest rate, house market, and the data set also has large deviation.

Outcome and Predictor Variables

Model Variables

```
# Define your outcome and predictor variables here
# Replace these placeholders with your real variables

outcome_var <- "Sale_Price"

predictor_vars <- c("Gr_Liv_Area", " Neighborhood", "Lot_Area", "Overall_Qual",

#outcome_var
#predictor_vars
```

Explanation:

- Outcome variable: Selling price of houses.
- Predictor variables: Neighborhood, Lot_Area, Overall_Qual, Garage_Cars, Gr_Liv_Area: Gross Living Area
- These elements are selected because they play a important role to influence house prices.

Data Cleaning and Preparation

Preparing the data

```r
# Example cleaning template
model_data <- bush_data %>%

     dplyr:: select(c(Sale_Price,
                      Gr_Liv_Area,
                      Garage_Cars,
                      Neighborhood,
                      Lot_Area,
                      Overall_Qual)) %>% drop_na()

summary(model_data)
```

```
  Sale_Price      Gr_Liv_Area    Garage_Cars                Neighborhood
Min.   : 12789   Min.   : 334   Min.   :0.000   North_Ames        : 443
1st Qu.:129500   1st Qu.:1126   1st Qu.:1.000   College_Creek     : 267
Median :160000   Median :1442   Median :2.000   Old_Town          : 239
Mean   :180796   Mean   :1500   Mean   :1.766   Edwards           : 194
3rd Qu.:213500   3rd Qu.:1743   3rd Qu.:2.000   Somerset          : 182
Max.   :755000   Max.   :5642   Max.   :5.000   Northridge_Heights: 166
                                                (Other)           :1439
   Lot_Area            Overall_Qual
Min.   :  1300   Average      :825
1st Qu.:  7440   Above_Average:732
Median :  9436   Good         :602
Mean   : 10148   Very_Good    :350
3rd Qu.: 11555   Below_Average:226
Max.   :215245   Excellent    :107
                 (Other)      : 88
```

Explanation:

- Any row having a column with NA data was removed by drop_na().
- No rows were filtered in the analysis but the significant elements were selected.

- One variable was created but wasn’t used for the lack of significance.

Exploratory Analysis

Visualizing the data

Gross Living Area:

```
ggplot(model_data, aes(x = Gr_Liv_Area, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explanation:

- This shows a positive relation between Gross Living Area and sale prices. More Gross Living Area tends give more value to the house prices.
- The outliers suggests there more other factors influencing the prices too.

Garage Cars:

```
ggplot(model_data, aes(x = Garage_Cars, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explanation:

- The blue line shows positive relation, as the number of capacity increases sales price goes up.

- The scattered data shows house prices are driven by some other factors too. There are some outliers.

  Neighborhood:

```
ggplot(model_data, aes(x = Neighborhood, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explanation:

- The graph shows different values in different neighborhoods and this is indication that neighborhoods play an important role in house pricing.

  Lot Area:

```
ggplot(model_data, aes(x = Lot_Area, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explanation:

- This graphs shows a positive but weak relation between lot area and sales price because the data is clustered in one place and too many outliers.

  Overall Quality:

```
ggplot(model_data, aes(x = Overall_Qual, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explanation:

- This graph shows different prices at different quality level.
- It indicates that the better the quality higher will be the price.

Linear Regression Model

Build the model

```r
# Create a model formula
 model_formula <- as.formula(
              paste("Sale_Price", " ~ ", paste(c(
                     "Gr_Liv_Area",
                     "Garage_Cars",
                     "Neighborhood",
                     "Lot_Area",
                     "Overall_Qual")), collapse = " + ") )

 model_formula
```

```
Sale_Price ~ Gr_Liv_Area + Sale_Price ~ Garage_Cars + Sale_Price ~
    Neighborhood + Sale_Price ~ Lot_Area + Sale_Price ~ Overall_Qual
```

Model 1: Neighborhood and Overall Quality

```r
# Fit the model
lm_model1 <- lm(Sale_Price ~ Neighborhood + Overall_Qual, data = model_data)

summary(lm_model1)
```

```
Call:
lm(formula = Sale_Price ~ Neighborhood + Overall_Qual, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-235002  -19935   -2523   16542  268477 

Coefficients:
                                                     Estimate Std. Error t value
(Intercept)                                             60276      19504   3.090
NeighborhoodCollege_Creek                               17078       3259   5.240
NeighborhoodOld_Town                                   -21305       3146  -6.772
NeighborhoodEdwards                                    -11815       3383  -3.492
NeighborhoodSomerset                                    17283       3888   4.445
```

```
NeighborhoodNorthridge_Heights                    53522       4458  12.005
NeighborhoodGilbert                               15711       3733   4.209
NeighborhoodSawyer                                -2088       3666  -0.569
NeighborhoodNorthwest_Ames                        18994       3993   4.757
NeighborhoodSawyer_West                           13830       4049   3.416
NeighborhoodMitchell                              10952       4070   2.691
NeighborhoodBrookside                            -16127       4168  -3.869
```

Brief explanation:

- This model is taking neighborhood and overall quality to predict the housing prices.
- It is giving ~0.77 R-square score which is a indication of a good model providing good accuracy.

Model 2: Overall Quality and Gross Living Area

```r
lm_model2 <- lm(Sale_Price ~ Overall_Qual + Gr_Liv_Area, data = model_data)

summary(lm_model2)
```

```
Call:
lm(formula = Sale_Price ~ Overall_Qual + Gr_Liv_Area, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-443413  -19743     182   18346  228183 

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)    
(Intercept)                 -172.24   18783.62  -0.009 0.992685    
Overall_QualPoor           16234.61   21414.04   0.758 0.448435    
Overall_QualFair           25475.71   19638.73   1.297 0.194659    
Overall_QualBelow_Average  43475.04   18893.03   2.301 0.021455 *  
Overall_QualAverage        65979.88   18778.45   3.514 0.000449 ***
Overall_QualAbove_Average  82762.85   18797.81   4.403 1.11e-05 ***
Overall_QualGood          113605.35   18831.17   6.033 1.81e-09 ***
Overall_QualVery_Good     167934.86   18903.89   8.884  < 2e-16 ***
Overall_QualExcellent     254166.12   19176.31  13.254  < 2e-16 ***
Overall_QualVery_Excellent 294564.71   20165.97  14.607  < 2e-16 ***
Gr_Liv_Area                    54.77       1.69  32.408  < 2e-16 ***
---
```

Brief explanation:

- This model is taking overall quality and Gross Living Area to predict the housing prices.
- It is giving ~0.78 R-square score which is a indication of a good model providing good accuracy.

Model 3: Lot Area and Garage Cars

```r
lm_model3 <- lm(Sale_Price ~ Lot_Area + Garage_Cars, data = model_data)

summary(lm_model3)
```

```
Call:
lm(formula = Sale_Price ~ Lot_Area + Garage_Cars, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-262240  -34732   -5710   24285  476025 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.994e+04  2.951e+03   16.92   <2e-16 ***
Lot_Area    1.575e+00  1.422e-01   11.07   <2e-16 ***
Garage_Cars 6.504e+04  1.472e+03   44.18   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 59660 on 2927 degrees of freedom
Multiple R-squared:  0.4427,	Adjusted R-squared:  0.4423 
F-statistic:  1163 on 2 and 2927 DF,  p-value: < 2.2e-16
```

Brief explanation:

- This model is taking lot area and garage cars to predict the housing prices.
- It is giving ~0.44 R-square score which is a indication of a weak model with low predictive accuracy.

Model Performance

How well does the model perform?

Model 1: Neighborhood and Overall Quality

```r
# Tidy model output
model_summary1 <- glance(lm_model1)
model_coefficients1 <- tidy(lm_model1)

model_summary1
```

```
# A tibble: 1 × 12
  r.squared adj.r.squared  sigma statistic p.value    df  logLik    AIC    BIC
      <dbl>         <dbl>  <dbl>     <dbl>   <dbl> <dbl>   <dbl>  <dbl>  <dbl>
1     0.769         0.766 38659.      267.       0    36 -35087. 70250. 70478.
# ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```r
model_coefficients1
```

```
# A tibble: 37 × 5
   term                          estimate std.error statistic  p.value
   <chr>                            <dbl>     <dbl>     <dbl>    <dbl>
 1 (Intercept)                     60276.    19504.     3.09  2.02e- 3
 2 NeighborhoodCollege_Creek       17078.     3259.     5.24  1.72e- 7
 3 NeighborhoodOld_Town           -21305.     3146.    -6.77  1.53e-11
 4 NeighborhoodEdwards            -11815.     3383.    -3.49  4.86e- 4
 5 NeighborhoodSomerset            17283.     3888.     4.45  9.11e- 6
 6 NeighborhoodNorthridge_Heights  53522.     4458.    12.0   1.94e-32
 7 NeighborhoodGilbert             15711.     3733.     4.21  2.65e- 5
 8 NeighborhoodSawyer              -2088.     3666.    -0.569 5.69e- 1
 9 NeighborhoodNorthwest_Ames      18994.     3993.     4.76  2.06e- 6
10 NeighborhoodSawyer_West         13830.     4049.     3.42  6.45e- 4
# ℹ 27 more rows
```

Discuss:

- The R-squared value is 0.7686918 or ~ 0.77.

- The model seems useful because the R-squared value is quiet big and the p- value is small.
- These values indicates that the model can predict the house prices by neighborhood and overall Quality with accuracy.

Model 2: Overall Quality and Gross Living Area

```
# Tidy model output
 model_summary2 <- glance(lm_model2)
 model_coefficients2 <- tidy(lm_model2)

 model_summary2
```

```
# A tibble: 1 × 12
  r.squared adj.r.squared  sigma statistic p.value    df  logLik    AIC    BIC
      <dbl>         <dbl>  <dbl>     <dbl>   <dbl> <dbl>   <dbl>  <dbl>  <dbl>
1     0.781         0.780 37446.     1041.       0    10 -35007. 70038. 70109.
# ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
 model_coefficients2
```

```
# A tibble: 11 × 5
   term                       estimate std.error statistic   p.value
   <chr>                         <dbl>     <dbl>     <dbl>     <dbl>
 1 (Intercept)                   -172.   18784.    -0.00917 9.93e-  1
 2 Overall_QualPoor            16235.   21414.     0.758   4.48e-  1
 3 Overall_QualFair            25476.   19639.     1.30    1.95e-  1
 4 Overall_QualBelow_Average   43475.   18893.     2.30    2.15e-  2
 5 Overall_QualAverage         65980.   18778.     3.51    4.49e-  4
 6 Overall_QualAbove_Average   82763.   18798.     4.40    1.11e-  5
 7 Overall_QualGood           113605.   18831.     6.03    1.81e-  9
 8 Overall_QualVery_Good      167935.   18904.     8.88    1.11e- 18
 9 Overall_QualExcellent      254166.   19176.    13.3     5.58e- 39
10 Overall_QualVery_Excellent 294565.   20166.    14.6     1.08e- 46
11 Gr_Liv_Area                    54.8      1.69  32.4     4.49e-197
```

Discuss:

- The R-squared value is 0.7810357 or ~ 0.78.

- The model seems useful because the R-squared value is quiet big and the p- value is small.
- These values indicates that the model can predict the house prices by Overall Quality and Gross Living Area with quiet accuracy.

Model 3: Lot Area and Garage Cars

```
# Tidy model output
 model_summary3 <- glance(lm_model3)
 model_coefficients3 <- tidy(lm_model3)

 model_summary3
```

```
# A tibble: 1 × 12
  r.squared adj.r.squared  sigma statistic p.value    df  logLik    AIC    BIC
      <dbl>         <dbl>  <dbl>     <dbl>   <dbl> <dbl>   <dbl>  <dbl>  <dbl>
1     0.443         0.442 59658.     1163.       0     2 -36375. 72759. 72783.
# ℹ 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

```
 model_coefficients3
```

```
# A tibble: 3 × 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept) 49940.    2951.         16.9 2.40e-61
2 Lot_Area        1.57     0.142      11.1 5.91e-28
3 Garage_Cars 65040.    1472.         44.2 0
```

Discuss:

- The R-squared value is 0.4426904 or ~ 0.44.
- The model seems weak and unreliable because the R-squared value is quiet small even though the p- value is small.

- These values indicates that the model can’t predict the house prices by lot area and garage cars with enough accuracy.

Most Important Variables

Which predictors matter most?

Model 1: Neighborhood and Overall Quality

```r
# View coefficients sorted by absolute value
model_coefficients1 %>%

  dplyr::filter(term != "(Intercept)") %>%

  dplyr::mutate(abs_estimate = abs(estimate)) %>%

  dplyr::arrange(desc(abs_estimate))
```

```
# A tibble: 36 × 6
   term                       estimate std.error statistic  p.value abs_estimate
   <chr>                         <dbl>     <dbl>     <dbl>    <dbl>        <dbl>
 1 Overall_QualVery_Excellent  346541.    20784.     16.7  1.14e-59      346541.
 2 Overall_QualExcellent       260938.    19990.     13.1  6.97e-38      260938.
 3 Overall_QualVery_Good       174451.    19646.      8.88 1.15e-18      174451.
 4 Overall_QualGood            126510.    19526.      6.48 1.08e-10      126510.
 5 Overall_QualAbove_Average    98062.    19464.      5.04 4.99e- 7       98062.
 6 NeighborhoodGreen_Hills      93214.    27479.      3.39 7.03e- 4       93214.
 7 NeighborhoodNorthridge       91143.     5506.     16.6  7.26e-59       91143.
 8 Overall_QualAverage          76748.    19444.      3.95 8.10e- 5       76748.
 9 Overall_QualBelow_Average    58796.    19561.      3.01 2.67e- 3       58796.
10 NeighborhoodStone_Brook      55084.     6398.      8.61 1.18e-17       55084.
# ℹ 26 more rows
```

Model 2: Overall Quality and Gross Living Area

```r
# View coefficients sorted by absolute value
model_coefficients2 %>%

  dplyr::filter(term != "(Intercept)") %>%

  dplyr::mutate(abs_estimate = abs(estimate)) %>%

  dplyr::arrange(desc(abs_estimate))
```

```
# A tibble: 10 × 6
   term                      estimate std.error statistic   p.value abs_estimate
   <chr>                        <dbl>     <dbl>     <dbl>     <dbl>        <dbl>
 1 Overall_QualVery_Excelle… 294565.   20166.      14.6   1.08e- 46     294565.
 2 Overall_QualExcellent     254166.   19176.      13.3   5.58e- 39     254166.
 3 Overall_QualVery_Good     167935.   18904.       8.88  1.11e- 18     167935.
 4 Overall_QualGood          113605.   18831.       6.03  1.81e-  9     113605.
 5 Overall_QualAbove_Average  82763.   18798.       4.40  1.11e-  5      82763.
 6 Overall_QualAverage        65980.   18778.       3.51  4.49e-  4      65980.
 7 Overall_QualBelow_Average  43475.   18893.       2.30  2.15e-  2      43475.
 8 Overall_QualFair           25476.   19639.       1.30  1.95e-  1      25476.
 9 Overall_QualPoor           16235.   21414.       0.758 4.48e-  1      16235.
10 Gr_Liv_Area                   54.8      1.69    32.4   4.49e-197         54.8
```

```r
# View coefficients sorted by absolute value
model_coefficients3 %>%

  dplyr::filter(term != "(Intercept)") %>%

  dplyr::mutate(abs_estimate = abs(estimate)) %>%

  dplyr::arrange(desc(abs_estimate))
```

```
# A tibble: 2 × 6
  term        estimate std.error statistic  p.value abs_estimate
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>        <dbl>
1 Garage_Cars 65040.    1472.         44.2 0            65040.
2 Lot_Area        1.57     0.142      11.1 5.91e-28         1.57
```

Explain:

- Neighborhood, Overall quality, and Gross Living Area appears to be the most important predictors.
- For neighborhoods there are no positive or negative effective because it depended on specific neighborhood. For overall quality, it has a positive effect on sale price and same goes for Gross Living Area.
- These insights matters because they reflect that the quality of structure and area or location matters when it comes to

housing prices.

Key Findings

Main conclusions from the models

```
Check <- model_data %>%

         dplyr:: select(Neighborhood, Sale_Price) %>%

         dplyr :: group_by(Neighborhood) %>%

         dplyr::summarise(total_price= sum(Sale_Price),

                          avg_price = mean(Sale_Price),

                          count = n() ) %>%

                          arrange(desc(avg_price))

Check
```

```
# A tibble: 28 × 4
   Neighborhood       total_price avg_price count
   <fct>                    <int>     <dbl> <int>
 1 Northridge            23452658   330319.    71
 2 Stone_Brook           16535689   324229.    51
 3 Northridge_Heights    53455032   322018.   166
 4 Green_Hills             560000   280000      2
 5 Veenker                5959550   248315.    24
 6 Timberland            17755167   246600.    72
 7 Somerset              41806733   229707.   182
 8 Clear_Creek            9181132   208662.    44
 9 Crawford              21377736   207551.   103
10 College_Creek         53881517   201803.   267
# ℹ 18 more rows
```

Claims:

- Model 1 and 2 predicts prices accurately using factors like neighborhood, overall quality, and Gross Living Area.

- A one-unit increase in overall excellent quality and Gross Living Area is associated with a ~$295K and ~$55 increase in housing price.
- Neighborhood and overall quality had the strongest estimated relationship with the sale price. Neighborhoods such as Northridge, Stone_Brook, and Northridge_Heights has over ~$320K avg sale price.
- The model 1 and 2 explained 77% and 78% of the variation in the outcome.

Basic Programming Concept 1: Variables

What is a variable?

```
# Example variables from this project

#outcome_var = "Sale price"

#predictor_vars = Neighborhoods, Overall_Qual, Lot_area, Garage_cars, Gr_Liv_Are
```

Explain in your own words: - A variable is in programming is a tag created where data such as quantitative, categorical, and identity are stored.

- We used variable to store specific function that were used to select the significant data, modify and present them in comprehensive way possible. Variables like Neighborhoods, Overall_Qual, Lot_area, Garage_cars, Gr_Liv_Area were used as predictor to predict the outcome variable which is Sales_Price.

Basic Programming Concept 2: Functions

What is a function?

```
# Example functions used in this project
 head(bush_data)
```

```
# A tibble: 6 × 81
  MS_SubClass             MS_Zoning Lot_Frontage Lot_Area Street Alley Lot_Shape
  <fct>                   <fct>            <dbl>    <int> <fct>  <fct> <fct>
1 One_Story_1946_and_New… Resident…          141    31770 Pave   No_A… Slightly…
2 One_Story_1946_and_New… Resident…           80    11622 Pave   No_A… Regular
3 One_Story_1946_and_New… Resident…           81    14267 Pave   No_A… Slightly…
4 One_Story_1946_and_New… Resident…           93    11160 Pave   No_A… Regular
5 Two_Story_1946_and_New… Resident…           74    13830 Pave   No_A… Slightly…
6 Two_Story_1946_and_New… Resident…           78     9978 Pave   No_A… Slightly…
# ℹ 74 more variables: Land_Contour <fct>, Utilities <fct>, Lot_Config <fct>,
#   Land_Slope <fct>, Neighborhood <fct>, Condition_1 <fct>, Condition_2 <fct>,
#   Bldg_Type <fct>, House_Style <fct>, Overall_Qual <fct>, Overall_Cond <fct>,
#   Year_Built <int>, Year_Remod_Add <int>, Roof_Style <fct>, Roof_Matl <fct>,
#   Exterior_1st <fct>, Exterior_2nd <fct>, Mas_Vnr_Type <fct>,
#   Mas_Vnr_Area <dbl>, Exter_Qual <fct>, Exter_Cond <fct>, Foundation <fct>,
#   Bsmt_Qual <fct>, Bsmt_Cond <fct>, Bsmt_Exposure <fct>, …
```

```
 glimpse(bush_data)
```

```
Rows: 2,930
Columns: 81
$ MS_SubClass        <fct> One_Story_1946_and_Newer_All_Styles, One_Story_1946…
$ MS_Zoning          <fct> Residential_Low_Density, Residential_High_Density, …
$ Lot_Frontage       <dbl> 141, 80, 81, 93, 74, 78, 41, 43, 39, 60, 75, 0, 63,…
$ Lot_Area           <int> 31770, 11622, 14267, 11160, 13830, 9978, 4920, 5005…
$ Street             <fct> Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pave, Pav…
$ Alley              <fct> No_Alley_Access, No_Alley_Access, No_Alley_Access, …
$ Lot_Shape          <fct> Slightly_Irregular, Regular, Slightly_Irregular, Re…
$ Land_Contour       <fct> Lvl, Lvl, Lvl, Lvl, Lvl, Lvl, Lvl, HLS, Lvl, Lvl, L…
$ Utilities          <fct> AllPub, AllPub, AllPub, AllPub, AllPub, AllPub, All…
$ Lot_Config         <fct> Corner, Inside, Corner, Corner, Inside, Inside, Ins…
$ Land_Slope         <fct> Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, Gtl, G…
$ Neighborhood       <fct> North_Ames, North_Ames, North_Ames, North_Ames, Gil…
$ Condition_1        <fct> Norm, Feedr, Norm, Norm, Norm, Norm, Norm, Norm, No…
```

```
$ Condition_2        <fct> Norm, Norm, Norm, Norm, Norm, Norm, Norm, Norm, Nor…
$ Bldg_Type          <fct> OneFam, OneFam, OneFam, OneFam, OneFam, OneFam, Twn…
$ House_Style        <fct> One_Story, One_Story, One_Story, One_Story, Two_Sto…
$ Overall_Qual       <fct> Above_Average, Average, Above_Average, Good, Averag…
$ Overall_Cond       <fct> Average, Above_Average, Above_Average, Average, Ave…
$ Year_Built         <int> 1960, 1961, 1958, 1968, 1997, 1998, 2001, 1992, 199…
$ Year_Remod_Add     <int> 1960, 1961, 1958, 1968, 1998, 1998, 2001, 1992, 199…
```

```r
#lm(model_formula, data = model_data)
summary(lm_model1)
```

```
Call:
lm(formula = Sale_Price ~ Neighborhood + Overall_Qual, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-235002  -19935   -2523   16542  268477 

Coefficients:
                                                    Estimate Std. Error t value
(Intercept)                                            60276      19504   3.090
NeighborhoodCollege_Creek                              17078       3259   5.240
NeighborhoodOld_Town                                  -21305       3146  -6.772
NeighborhoodEdwards                                   -11815       3383  -3.492
NeighborhoodSomerset                                   17283       3888   4.445
NeighborhoodNorthridge_Heights                         53522       4458  12.005
NeighborhoodGilbert                                    15711       3733   4.209
NeighborhoodSawyer                                     -2088       3666  -0.569
NeighborhoodNorthwest_Ames                             18994       3993   4.757
NeighborhoodSawyer_West                                13830       4049   3.416
NeighborhoodMitchell                                   10952       4070   2.691
NeighborhoodBrookside                                 -16127       4168  -3.869
```

```r
summary(lm_model2)
```

```
Call:
lm(formula = Sale_Price ~ Overall_Qual + Gr_Liv_Area, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-443413  -19743     182   18346  228183 

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                 -172.24   18783.62  -0.009 0.992685
Overall_QualPoor           16234.61   21414.04   0.758 0.448435
Overall_QualFair           25475.71   19638.73   1.297 0.194659
```

```
Overall_QualBelow_Average   43475.04   18893.03   2.301 0.021455 *
Overall_QualAverage         65979.88   18778.45   3.514 0.000449 ***
Overall_QualAbove_Average   82762.85   18797.81   4.403 1.11e-05 ***
Overall_QualGood           113605.35   18831.17   6.033 1.81e-09 ***
Overall_QualVery_Good      167934.86   18903.89   8.884  < 2e-16 ***
Overall_QualExcellent      254166.12   19176.31  13.254  < 2e-16 ***
Overall_QualVery_Excellent 294564.71   20165.97  14.607  < 2e-16 ***
Gr_Liv_Area                    54.77       1.69  32.408  < 2e-16 ***
---
```

```
summary(lm_model3)
```

```
Call:
lm(formula = Sale_Price ~ Lot_Area + Garage_Cars, data = model_data)

Residuals:
    Min      1Q  Median      3Q     Max 
-262240  -34732   -5710   24285  476025 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 4.994e+04  2.951e+03   16.92   <2e-16 ***
Lot_Area    1.575e+00  1.422e-01   11.07   <2e-16 ***
Garage_Cars 6.504e+04  1.472e+03   44.18   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 59660 on 2927 degrees of freedom
Multiple R-squared:  0.4427,	Adjusted R-squared:  0.4423 
F-statistic:  1163 on 2 and 2927 DF,  p-value: < 2.2e-16
```

Explain in your own words:

- A function is a code that does a specific task.
- A function that I used was summary(lm_model1).
- It takes model 1 as input which consists the data of Neighborhoods and Overall_Qual.

- As output it summarises the model and gives us R-square, residual, and p-value etc. to see if the model is reliable or not..

Basic Programming Concept 3: Data Types

What data types appear in my project?

```r
# Check data types
sapply(model_data, class)
```

```
Sale_Price  Gr_Liv_Area  Garage_Cars Neighborhood     Lot_Area Overall_Qual
 "integer"    "integer"    "numeric"     "factor"    "integer"     "factor"
```

Explain:

- We have found integers, numeric, and factor type data.
- Data types matter because R interprets different data types differently. To do calculations and plotting specific graphs, it uses numeric values but for grouping or for comparison it uses factor values.
- They effect the analysis a lot because if a data type is not represented correctly the model might collapse or miss interpret the task.

Advanced Concept 1: Pareto Principle

Applying the Pareto Principle

```
# Optional: create a simple ranking of predictor importance
 model_coefficients1 %>%
  dplyr:: filter(term != "(Intercept)") %>%
  dplyr::mutate(abs_estimate = abs(estimate)) %>%
  dplyr::arrange(desc(abs_estimate))
```

```
# A tibble: 36 × 6
   term                        estimate std.error statistic  p.value abs_estimate
   <chr>                          <dbl>     <dbl>     <dbl>    <dbl>        <dbl>
 1 Overall_QualVery_Excellent   346541.    20784.      16.7  1.14e-59      346541.
 2 Overall_QualExcellent        260938.    19990.      13.1  6.97e-38      260938.
 3 Overall_QualVery_Good        174451.    19646.       8.88 1.15e-18      174451.
 4 Overall_QualGood             126510.    19526.       6.48 1.08e-10      126510.
 5 Overall_QualAbove_Average     98062.    19464.       5.04 4.99e- 7       98062.
 6 NeighborhoodGreen_Hills       93214.    27479.       3.39 7.03e- 4       93214.
 7 NeighborhoodNorthridge        91143.     5506.      16.6  7.26e-59       91143.
 8 Overall_QualAverage           76748.    19444.       3.95 8.10e- 5       76748.
 9 Overall_QualBelow_Average     58796.    19561.       3.01 2.67e- 3       58796.
10 NeighborhoodStone_Brook       55084.     6398.       8.61 1.18e-17       55084.
# ℹ 26 more rows
```

Explain in your own words:

- The Pareto principle means a small number of factor or factors make up most of the outcome value.
- In this project among all the elements only Neighborhoods, Overall_Qual, Gr_living_Area had more impact.
- House prices are driven by some factors.

Advanced Concept 2: Correlation

Understanding relationships between variables

```r
# Keep only numeric columns for correlation analysis
numeric_data <- model_data %>%

dplyr:: select(where(is.numeric))

 cor(numeric_data, use = "complete.obs")
```

```
            Sale_Price Gr_Liv_Area Garage_Cars  Lot_Area
Sale_Price   1.0000000   0.7067799   0.6475616 0.2665492
Gr_Liv_Area  0.7067799   1.0000000   0.4878292 0.2855992
Garage_Cars  0.6475616   0.4878292   1.0000000 0.1794561
Lot_Area     0.2665492   0.2855992   0.1794561 1.0000000
```

```r
# Optional: visualize a key relationship
ggplot(model_data, aes(x = Gr_Liv_Area, y = Sale_Price)) +

  geom_point() +

  geom_smooth(method = "lm", se = FALSE)
```

Explain:

- Correlation means how two variables relate to each other.
- In this project, Gross Living Area has a correlation.
- It has a strong and positive correlation.

Advanced Concept 3: Standard Normal Form

Standardizing a variable with z-scores

```
# Example: standardize the outcome variable if it is numeric
# Replace with your actual numeric variable name if needed

model_data <- model_data %>%

   mutate(outcome_z = as.numeric(scale(Sale_Price)))

 summary(model_data$outcome_z)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
-2.1031 -0.6421 -0.2603  0.0000  0.4094  7.1877 
```

Explain:

- z-score shows how far the value is from actual mean and it is measured in standard deviations.
- z-scores or standardization helps with comparing values and it puts every data on a same scale.
- z-score shows how many standard deviation away is the actual value is from mean.

Recommendations

What should the audience do with these findings?

- For investments and later resell values, property investors and home buyers should focus on Neighborhoods, the quality of the houses, and bigger living area. It ensures best possible investment to for future resell value for every individual and entity.

Final Takeaways

Conclusion

Summarize:

- We predicted the sale prices of houses.
- The factors mattered most was overall quality, neighborhood, Gross Living Area.
- Both of our model 1 and 2 gave us good R-square such as 0.77 and 0.78. These values suggest these models are quiet accurate.
- I learned how we can use code to analyze vast amount by using some function and get the mean, sum. We can make different graphs, correlation, and pareto to visualize text data.

References or Appendix

Ames Hosusing ### Optional
Include citations, extra charts, or supporting output if needed.

Speaker notes